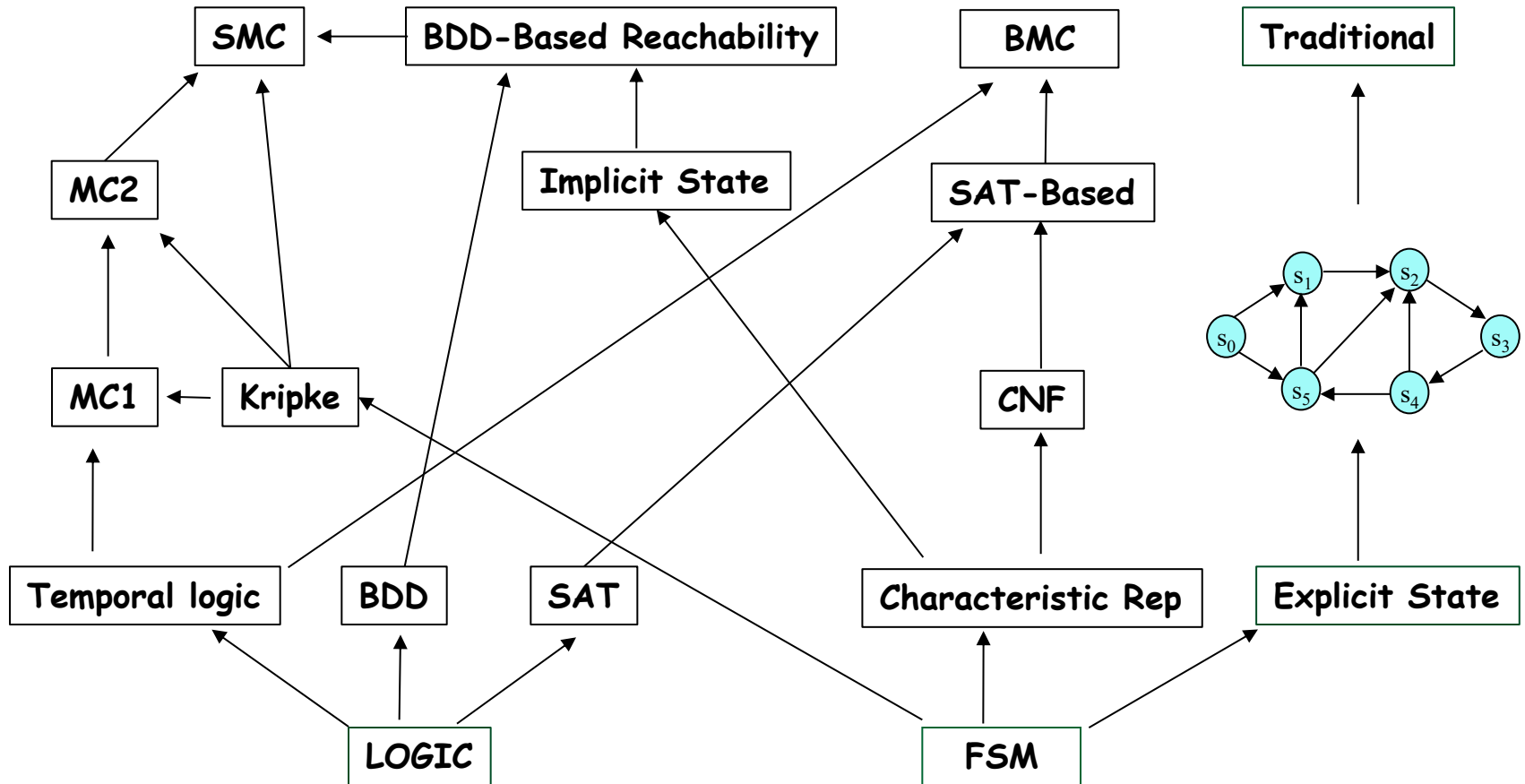




SAT-based Verification



A Road Map



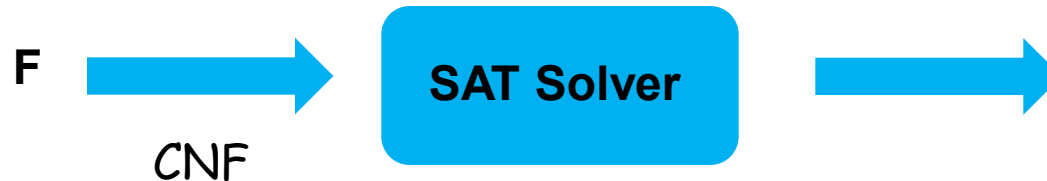
Satisfiability (SAT) Solver



$$f = (a + c) (b + c) (\neg a + \neg b + \neg c)$$

- Given a propositional formula in CNF, find an assignment to boolean variables that makes the formula true!
- The solver reports
 - if F is satisfiable or unsatisfiable.
 - if satisfiable, at least one assignment is reported.
 - Almost all modern SAT solver operates on CNF

Solving Problems with SAT Solvers



- Formulate the problem logic for SAT solver
- Convert your problem into the input format of a SAT solver
 - If the SAT solver requires a CNF form, convert it into a CNF form.
 - Convert the CNF form into the input numeric form.
- Run the solver and get the answer.
- Interpret the answer
 - Unsatisfiable: Meaning?
 - Satisfiable: Meaning? Model instances ?

CNF for Characteristic Representation (CNF_CR)

- Consider an inverter:
 - $y = \neg x$
 - $CR = y \Leftrightarrow \neg x$
 - $= (y \rightarrow \neg x) \wedge (\neg x \rightarrow y)$
 - $= (\neg y \vee \neg x) \wedge (x \vee y)$
 - CNF_CR

CNF for Characteristic Representation

- Consider an AND gate:
 - $c = ab$
 - $CR = c \Leftrightarrow ab$
 - $= (c \rightarrow ab) \wedge (ab \rightarrow c)$
 - $= (\neg c \vee ab) \wedge (\neg a \vee \neg b \vee c)$
 - $= (\neg c \vee a) \wedge (\neg c \vee b) \wedge (\neg a \vee \neg b \vee c)$
 - CNF_CR

CNFs for Characteristic Representations

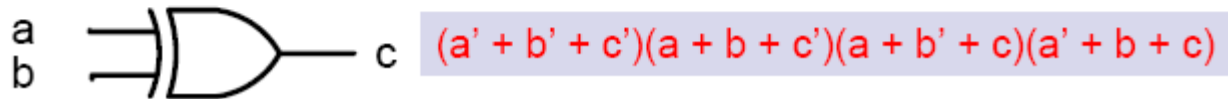
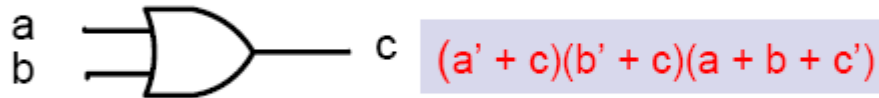
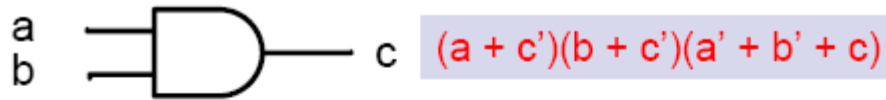
Negation: $x \leftrightarrow \bar{y} \Leftrightarrow (x \rightarrow \bar{y}) \wedge (\bar{y} \rightarrow x)$
 $\Leftrightarrow (\bar{x} \vee \bar{y}) \wedge (y \vee x)$

Disjunction: $x \leftrightarrow (y \vee z) \Leftrightarrow (y \rightarrow x) \wedge (z \rightarrow x) \wedge (x \rightarrow (y \vee z))$
 $\Leftrightarrow (\bar{y} \vee x) \wedge (\bar{z} \vee x) \wedge (\bar{x} \vee y \vee z)$

Conjunction: $x \leftrightarrow (y \wedge z) \Leftrightarrow (x \rightarrow y) \wedge (x \rightarrow z) \wedge ((y \wedge z) \rightarrow x)$
 $\Leftrightarrow (\bar{x} \vee y) \wedge (\bar{x} \vee z) \wedge ((y \wedge z) \vee x)$
 $\Leftrightarrow (\bar{x} \vee y) \wedge (\bar{x} \vee z) \wedge (\bar{y} \vee \bar{z} \vee x)$

Equivalence: $x \leftrightarrow (y \leftrightarrow z) \Leftrightarrow (x \rightarrow (y \leftrightarrow z)) \wedge ((y \leftrightarrow z) \rightarrow x)$
 $\Leftrightarrow (x \rightarrow ((y \rightarrow z) \wedge (z \rightarrow y))) \wedge ((y \leftrightarrow z) \rightarrow x)$
 $\Leftrightarrow (x \rightarrow (y \rightarrow z)) \wedge (x \rightarrow (z \rightarrow y)) \wedge ((y \leftrightarrow z) \rightarrow x)$
 $\Leftrightarrow (\bar{x} \vee \bar{y} \vee z) \wedge (\bar{x} \vee \bar{z} \vee y) \wedge ((y \leftrightarrow z) \rightarrow x)$
 $\Leftrightarrow (\bar{x} \vee \bar{y} \vee z) \wedge (\bar{x} \vee \bar{z} \vee y) \wedge (((y \wedge z) \vee (\bar{y} \wedge \bar{z})) \rightarrow x)$
 $\Leftrightarrow (\bar{x} \vee \bar{y} \vee z) \wedge (\bar{x} \vee \bar{z} \vee y) \wedge ((y \wedge z) \rightarrow x) \wedge ((\bar{y} \wedge \bar{z}) \rightarrow x)$
 $\Leftrightarrow (\bar{x} \vee \bar{y} \vee z) \wedge (\bar{x} \vee \bar{z} \vee y) \wedge (\bar{y} \vee \bar{z} \vee x) \wedge (y \vee z \vee x)$

Gates in CNF_CR



- $CNF_CR(\dots)$ = Characteristic function in CNF (...)

CNF for Characteristic Representation

- $c = ab$
 - $c \Leftrightarrow ab$
 - $= (c \rightarrow ab) \wedge (ab \rightarrow c)$
 - $= (\neg c \vee ab) \wedge (\neg a \vee \neg b \vee c)$
 - $= (\neg c \vee a) \wedge (\neg c \vee b) \wedge (\neg a \vee \neg b \vee c)$
- $e = abcd$
 - $e \Leftrightarrow abcd$
 - $= (e \rightarrow abcd) \wedge (abcd \rightarrow e)$
 - $= (\neg e \vee abcd) \wedge (\neg(abcd) \vee e)$
 - $= (\neg e \vee ab) \wedge (\neg e \vee cd) \wedge (\neg(abcd) \vee e)$
 - $= (\neg e \vee a) \wedge (\neg e \vee b) \wedge (\neg e \vee c) \wedge (\neg e \vee d) \wedge (\neg a \vee \neg b \vee \neg c \vee \neg d \vee e)$

CNF = POS ?

Minterm Index	A	B	C	F
0	0	0	0	0
1	0	0	1	0
2	0	1	0	0
3	0	1	1	1
4	1	0	0	0
5	1	0	1	1
6	1	1	0	1
7	1	1	1	1

$$F(A,B,C) = \bar{A}BC + A\bar{B}C + AB\bar{C} + ABC$$
$$= m_3 + m_5 + m_6 + m_7 = \sum(3,5,6,7)$$

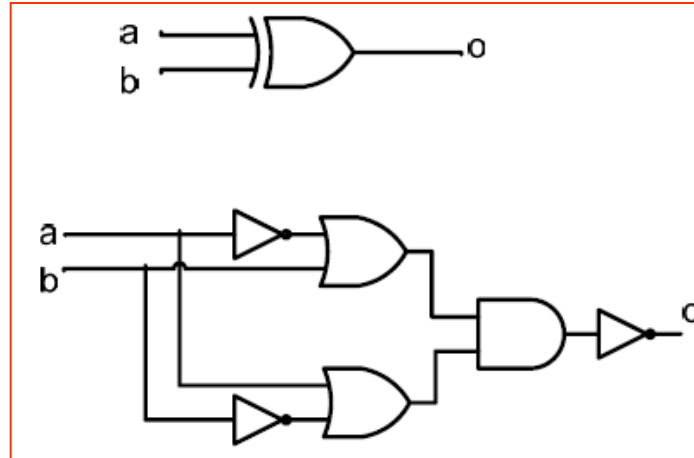
$$F(A,B,C) = (A+B+C)(A+B+\bar{C})(A+\bar{B}+C)(\bar{A}+B+C)$$
$$= M_0 \cdot M_1 \cdot M_2 \cdot M_4 = \prod(0,1,2,4)$$

Any Formula into CNF: Tseitin's Encoding

- It is possible to convert an arbitrary function into CNF
- Without introducing new variables, the size of the resulting formula will grow **exponentially**. Not practical.
- By introducing intermediate variables, the size of the resulting formula can grow **linearly**.
- The original and converted formulas are "equi-satisfiable".
- This technique is called Tseitin's encoding.

- $M = (A \rightarrow (B \wedge C)) = (a_1 \leftrightarrow (A \rightarrow a_2)) \wedge (a_2 \leftrightarrow (B \wedge C)) \wedge (a_1)$
- $(a_1 \leftrightarrow (A \rightarrow a_2)) = (a_1 \vee A) \wedge (a_1 \vee \neg a_2) \wedge (\neg a_1 \vee \neg A \vee a_2)$
- $(a_2 \leftrightarrow (B \wedge C)) = (\neg a_2 \vee B) \wedge (\neg a_2 \vee C) \wedge (a_2 \vee \neg B \vee \neg C)$

Equivalence Checking as Validity Problem



- $C_1 = C_2$ means
 - for all values for inputs of C_1 and C_2 , we have $o_1 = o_2$.
 - The equivalence problem is a **validity** problem:
 - $\forall I. [o_1 \Leftrightarrow o_2]_I$

Validity Problem by SAT Solvers

Validity: $\forall I. F=1$



SAT solver reports if F is satisfiable or **unsatisfiable**.



Valid formulas	Satisfiable, but non-valid formula	Unsatisfiable formulas
G	$F < \text{---} > \neg F$	$\neg G$

Negation of Validity Problems

Validity Problem G

x	y	G
0	1	1
1	0	1
0	0	1
1	1	1

Negation of G



Unsatisfiable

x	y	$\neg G$
0	1	0
1	0	0
0	0	0
1	1	0

x	y	G
0	1	0
1	0	1
0	0	1
1	1	1

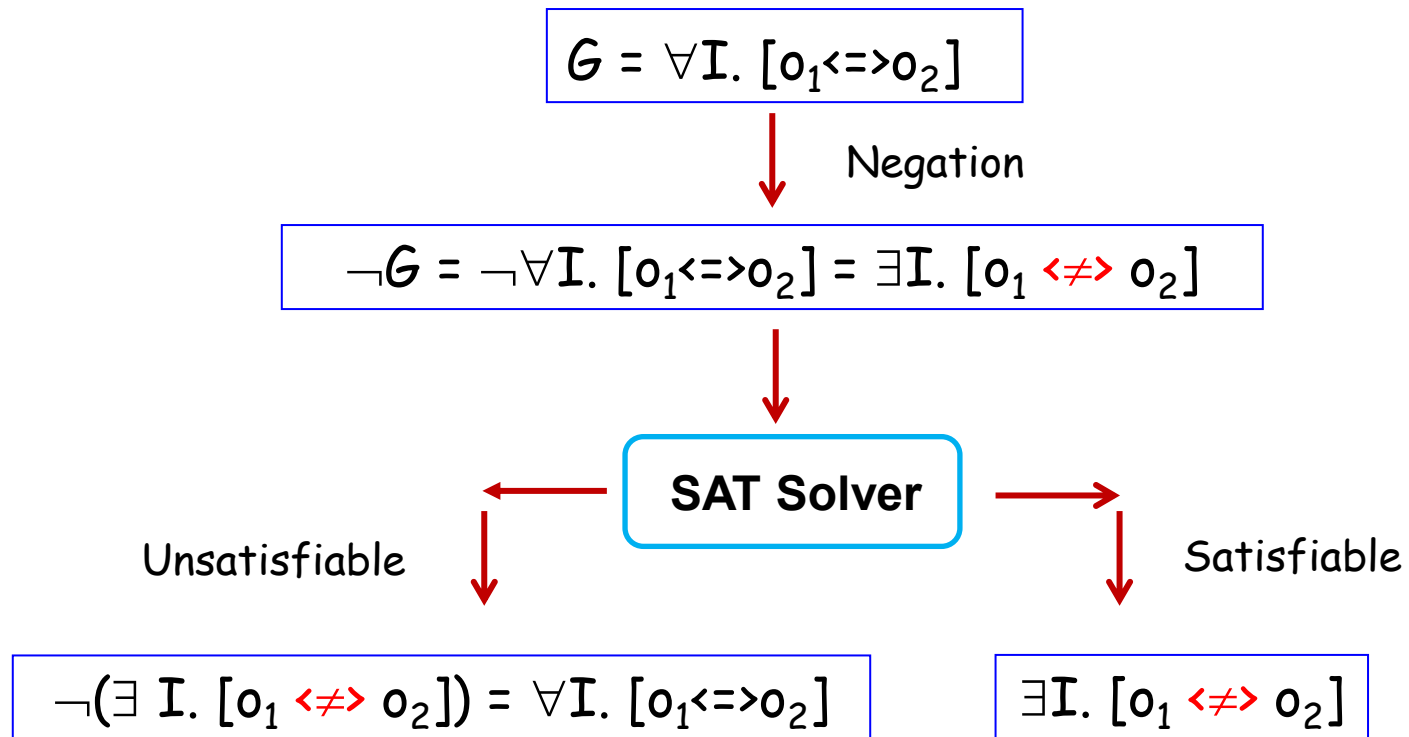
Negation of G



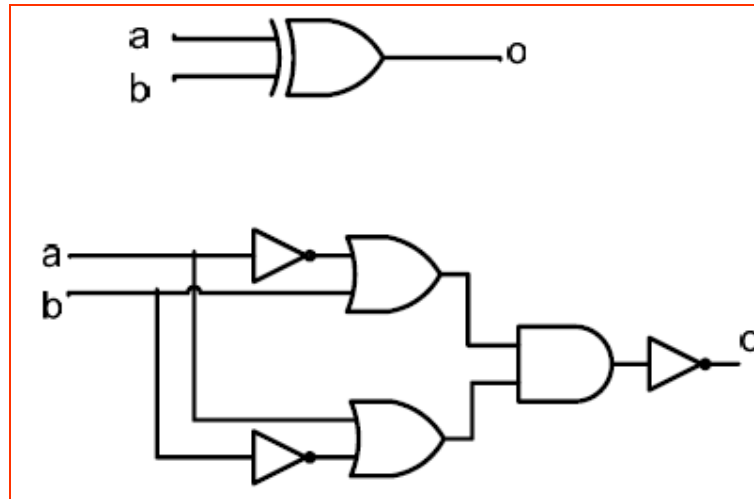
x	y	$\neg G$
0	1	1
1	0	0
0	0	0
1	1	0

Satisfiable

Validity $\rightarrow$ Unsatisfiability

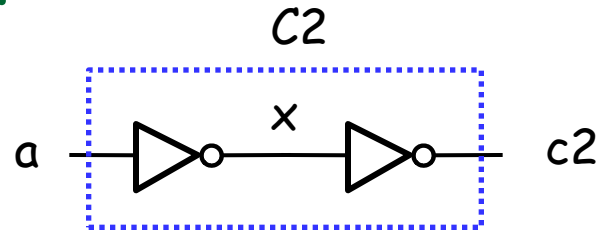
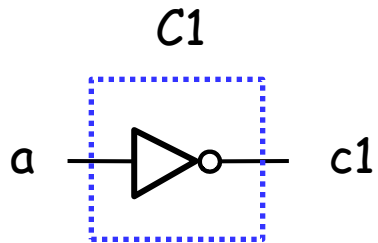


Negation of Equivalence



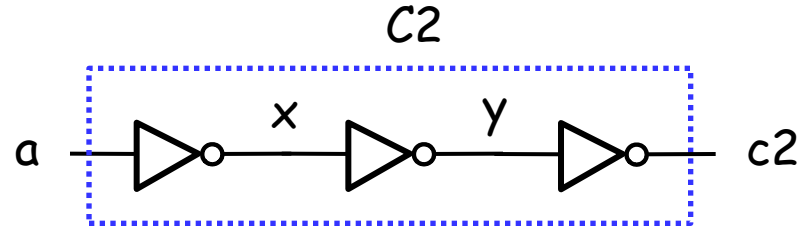
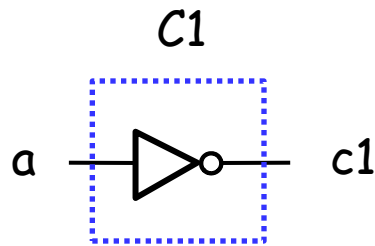
- $CNF_CR(C_1) \wedge CNF_CR(C_2) \wedge CNF [\neg(o_1 \leftrightarrow o_2)]$
 - $= CNF_CR(C_1) \wedge CNF_CR(C_2) \wedge (o_1 + o_2) \wedge (\neg o_1 + \neg o_2)$
- $CNF_CR(\dots)$ = Characteristic function in CNF (...)

Example 1



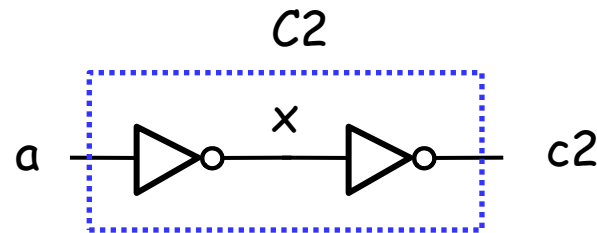
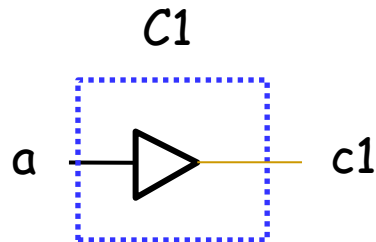
- Equivalence: $\forall a. c1 \Leftrightarrow c2$ (validity problem)
- Negation of the problem: $\exists a. c1 \not\Leftrightarrow c2$ ($= c1 \neg c2 + \neg c1 c2$)
- The problem for satisfiability:
 - $CR(C1) \wedge CR(C2) \wedge (c1 \neg c2 + \neg c1 c2)$
 - $= (a \neg c1 + \neg a c1) \wedge [(a \neg x + \neg a x) \wedge (x \neg c2 + \neg x c2)] \wedge (c1 \neg c2 + \neg c1 c2)$
 - $= \neg a x c1 \neg c2 + a \neg x \neg c1 c2$
 - Satisfiable: $(a=0, x=1, c1=1, c2=0); (a=1, x=0, c1=0, c2=1).$

Example 2



- Equivalence: $\forall a. c1 \Leftrightarrow c2$ (validity problem)
- Negation of the problem: $\exists a. c1 \not\Leftrightarrow c2$ ($= c1 \neg c2 + \neg c1 c2$)
- The problem for satisfiability:
 - $CR(C1) \wedge CR(C2) \wedge (c1 \neg c2 + \neg c1 c2)$
 - $= (a \neg c1 + \neg a c1) \wedge [(a \neg x + \neg a x) \wedge (x \neg y + \neg x y) \wedge (y \neg c2 + \neg y c2)] \wedge (c1 \neg c2 + \neg c1 c2)$
 - $= 0$
 - Unsatisfiable !
- Conclusion: $\forall a. c1 \not\Leftrightarrow c2$

Example: Buffer in CNF Forms



- Equivalence: $\forall a. c1 \Leftrightarrow c2$
- Negation of the problem: $\exists a. c1 \not\Leftrightarrow c2 = c1c2' + c1'c2 = (c1+c2)(c1'+c2')$
- $CNF_CR(C1) = a \Leftrightarrow c1 = (a'+c1)(a+c1')$
- $CNF_CR(C2) = \exists x. (a' \Leftrightarrow x) (a' \Leftrightarrow c2) = (a+c2')(a'+c2)$
- Input of SAT solver = $CNF_CR(C1) \wedge CNF_CR(C2) \wedge (c1+c2)(c1'+c2')$
 - $= (a'+c1)(a+c1') (a+c2')(a'+c2) (c1+c2)(c1'+c2')$
 - $= 0$ (Unsatisfiable)
- Conclusion: $\forall a. c1 \Leftrightarrow c2$

References

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- William K. Lam, "Hardware Design Verification: Simulation and Formal Method-Based Approaches", Prentice Hall, 2005.